

Solution Requirements

Date: 15 March 2026 **Team ID:** SWTID-2026-1506

Project Name: AI StudyBuddy

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register and log in using their email and password.	High
2	Authorization levels	Students can access study features; authorized users can access the functions permitted to their role.	High
3	External interfaces	The system connects to MongoDB for data storage and uses the required AI service/API for AI-based study assistance.	High
4	Transactions processing	The system processes user registration, login, study requests, and stores/retrieves related data.	High
5	Reporting	The system displays study-related information and user activity/results through the application interface.	Medium
6	Business rules, etc.	Only registered users can access protected study features. Required fields must be completed before submitting requests.	High
7	Compliance to laws or regulations	User information should be handled securely and only used for the purposes of the application.	Medium
8	Other	The system provides AI-assisted study support through the web application.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages and normal user actions should respond quickly.	Normal page response within 2 seconds under typical load.
2	Scalability	The application should support growth in users and stored study data.	Should support increasing users/data without major redesign.
3	Security & Data Privacy	User credentials and personal data must be protected; passwords should not be stored as plain text.	Passwords securely hashed; protected data accessible only to authorized users.
4	Reliability & Availability	The application should continue operating reliably and handle common errors without crashing.	Application remains available during normal operation and shows clear error messages.
5	Usability & Accessibility	The interface should be simple and easy for students to use on common devices.	Users should be able to complete main tasks with clear navigation and readable UI.
6	Other	Maintainability and portability should be considered for future updates and deployment.	Code should be organized into manageable components and run in the intended development/deployment environment.